

Method categories and comparator framework

a Transfer assumptions

1



Foundation models

Reusable cell-state representation
scGPT, scFoundation, scBERT

2



Latent / compositional models

Latent perturbation effect
scGen, CPA

3



Prior-graph models

Regulatory wiring
GEARS, AttentionPert

4



Optimal-transport models

Distributional intervention
CellIOT, scPRAM

5



Hybrid models

Representation + transition operator
STATE, PertAdapt

6



Chemistry-aware models

Compound-structure signal
chemCPA, FP-ridge

b Benchmark roles

1



Conditioned predictors

General predictive models
evaluated on held-out tasks

2



Task-specific deterministic predictors

FP-ridge, linear-shift-KOemb

3



Distributional comparator

CINEMA-OT

*reference only; never counted
as a conditioned win*

4

Baselines

Universal floor



cell-mean shift
linear-PCA shift

Context baselines



control-as-prediction
donor shift
training-mean shift

c Decision rule

Conditioned prediction



Compare against



cell-mean shift

AND



linear-PCA shift



Counted as working

only if it beats both
floor members



Context baselines and
task-specific comparators are
reported for interpretation



CINEMA-OT

distributional reference only